Exercise: Updating services and maintaining application health in Docker Compose

In this hands-on exercise, you'll **simulate how to manage and maintain a multi-service application using Docker Compose**. You will update a running service, implement a health check, and practice rolling back changes in case of failure. This will help you understand the practical steps for ensuring application reliability and continuity in production environments.

Learning objective

> - Be able to gain insights into updating services and maintaining the health of the entire application stack.

Instructions

Step 1: Update a service in Docker Compose

> Imagine you're running a multi-service app using Docker Compose. One of the services has a new image version available. Your task is to:
>
> - Open the `docker-compose.yml` file.
> - Update the `image` tag of one service to a newer version (e.g., `nginx:1.25`).
> - Apply the update by running `docker-compose up -d`.
>
> **Checkpoint:** Use `docker-compose ps` to confirm the updated container is running with the new image.

Step 2: Add a health check

> To monitor your updated service, configure a `healthcheck` in your `docker-compose.yml`:
>
> - Add a health check command to test if the service is responsive (e.g., using `curl` to check a web server).
> - Set the interval, timeout, and retry settings.
>
> **Example:**

```
services:
  web:
    image: nginx:1.25
    healthcheck:
      test: ["CMD", "curl", "-f", "http://localhost"]
      interval: 30s
      timeout: 10s
      retries: 3
```

Checkpoint: Run `docker-compose ps` and confirm the health status says `(healthy)`.

Step 3: Simulate a rollback ✖

Let's say the new image you pulled caused unexpected errors. Roll back the service to the previous version:

- Edit the `image` tag in `docker-compose.yml` to the previous version (e.g., `nginx:1.24`).
- Recreate the container by running `docker-compose up -d --force-recreate`.

Checkpoint: Use `docker-compose logs <service>` to verify the rollback succeeded.

Deliverables

- **Updated docker-compose.yml:** Includes a modified image and a configured healthcheck.
- **Terminal output:**
  - Screenshot or copy of `docker-compose ps` showing updated and healthy service.
  - Logs confirming service restart or rollback.
- **Short summary:** Write 2–3 sentences explaining what changes you made and what you learned about managing services in Docker Compose.